1ST Qatar Conference on Laboratory Medicine and Pathology

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Ethical considerations in the use of AI in Healthcare

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Learning Objectives

To equip healthcare professionals with the ethical awareness and practical frameworks needed to use AI responsibly, balancing innovation with patient safety, equity, and human oversight.

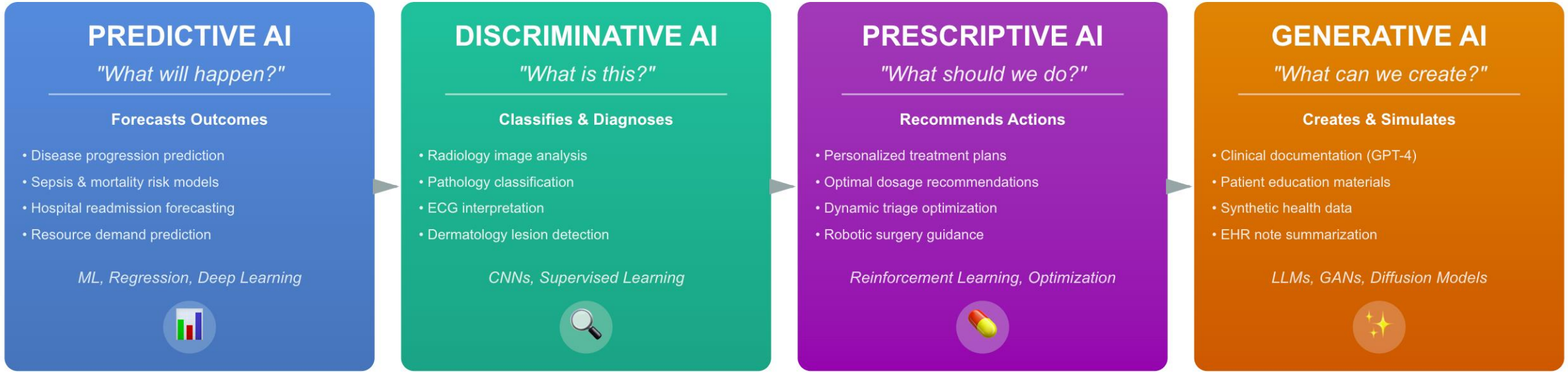
Disclosure/ Caveats

Reviewed for compliance with the MoPH-DHP accreditation standards.

No Conflict of Interest: either financial or non-financial with profit or not-for-profit organizations.

I am a Computing/ Engineering Professor, not a Physician nor a Pathologist, but I do have work in AI Ethics and Healthcare AI

Typology of AI: *What it is and how it is used?*



REPLACEMENT

Automation or Autonomous Action

AUGMENTATION

AI in a Supporting Role

THE ETHICAL CASE: WHY USE TECHNOLOGY?

10–20% of all health cases delayed, missed or incorrectly diagnosed

(Graber et al., Journal of the American Medical Association (JAMA), 2012)

HUMAN FACTORS RESEARCH:
"Systems that rely on error-free performance are doomed to fail."
— Lucian L. Leape

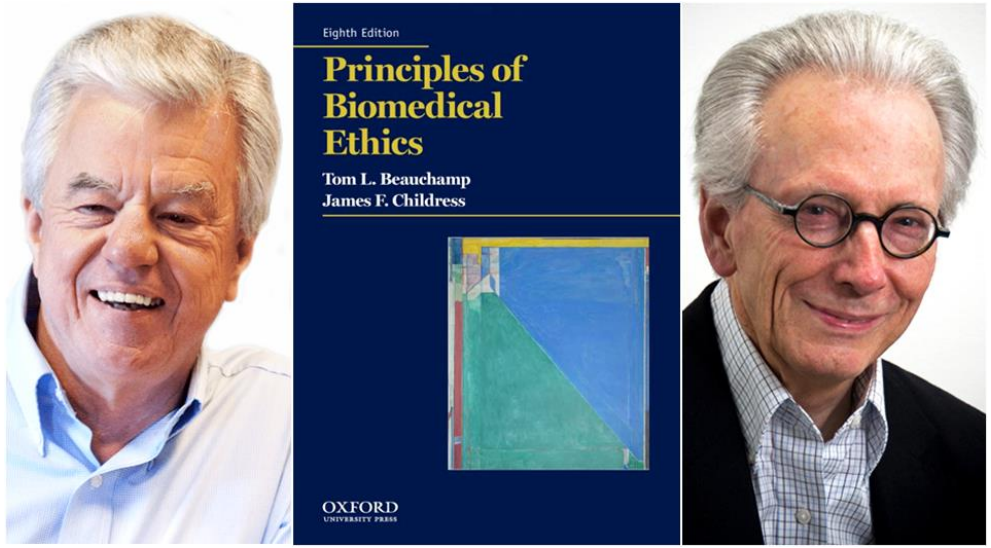
Various imbalances:

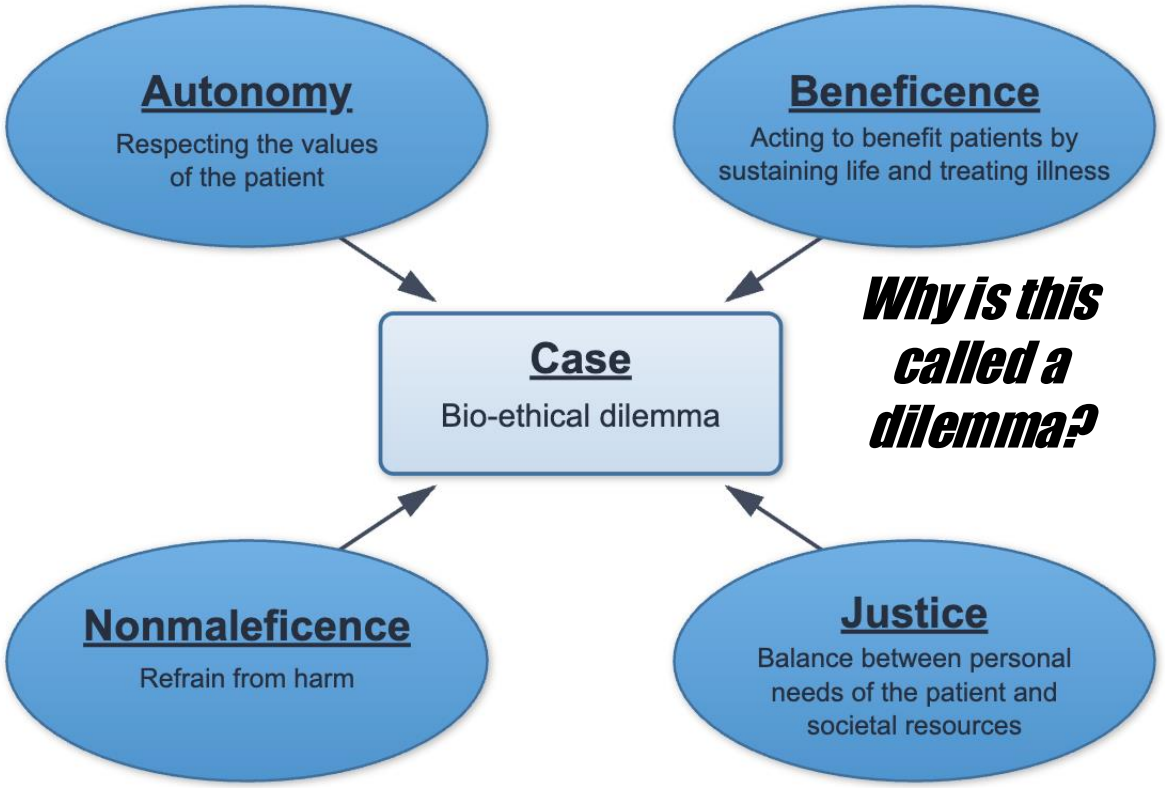
- Geographical *(Urban vs. Rural)*
- Specialty based
- Gender based

The Evolution of Medical Morality and Bioethics



Do No Harm
Act for the Patient's Benefit
Respect Life and Confidentiality
Maintain Professional Integrity
Teach and Preserve the Art
Respect Colleagues and Teachers

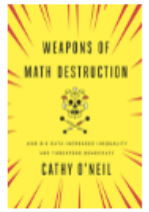




The principles often ***conflict with one another****, forcing us to balance equally valid but competing moral obligations rather than choose a single "right" answer.*

The Landscape of AI Ethics

> THERE ARE ETHICAL CHOICES IN EVERY SINGLE ALGORITHM WE BUILD



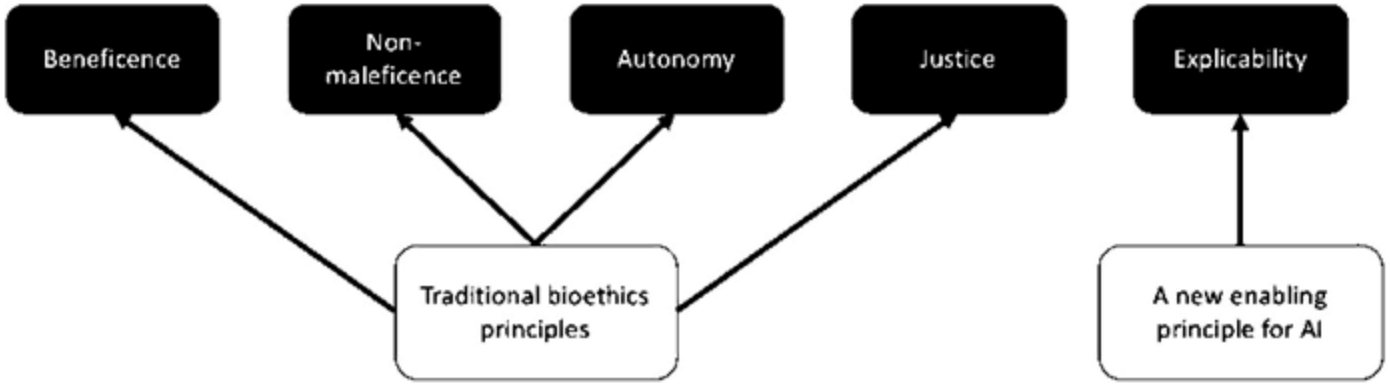
nature machine intelligence
PERSPECTIVE
https://doi.org/10.1038/s42256-019-0088-2

The global landscape of AI ethics guidelines

Anna Jobin, Marcello Ienca and Effy Vayena*



Four higher-level ethical principles—beneficence, non-maleficence, justice, and autonomy—capture the essence of most AI declarations



nature machine intelligence
PERSPECTIVE

Principles alone cannot guarantee ethical AI

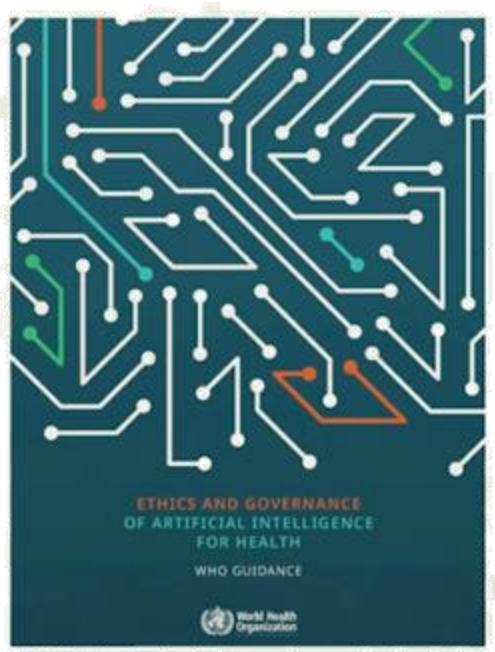
Brent Mittelstadt

A multipronged approach is needed combining public policy, regulation, economic incentives, accountability as well as technical research.



WHO consensus ethical healthcare AI principles

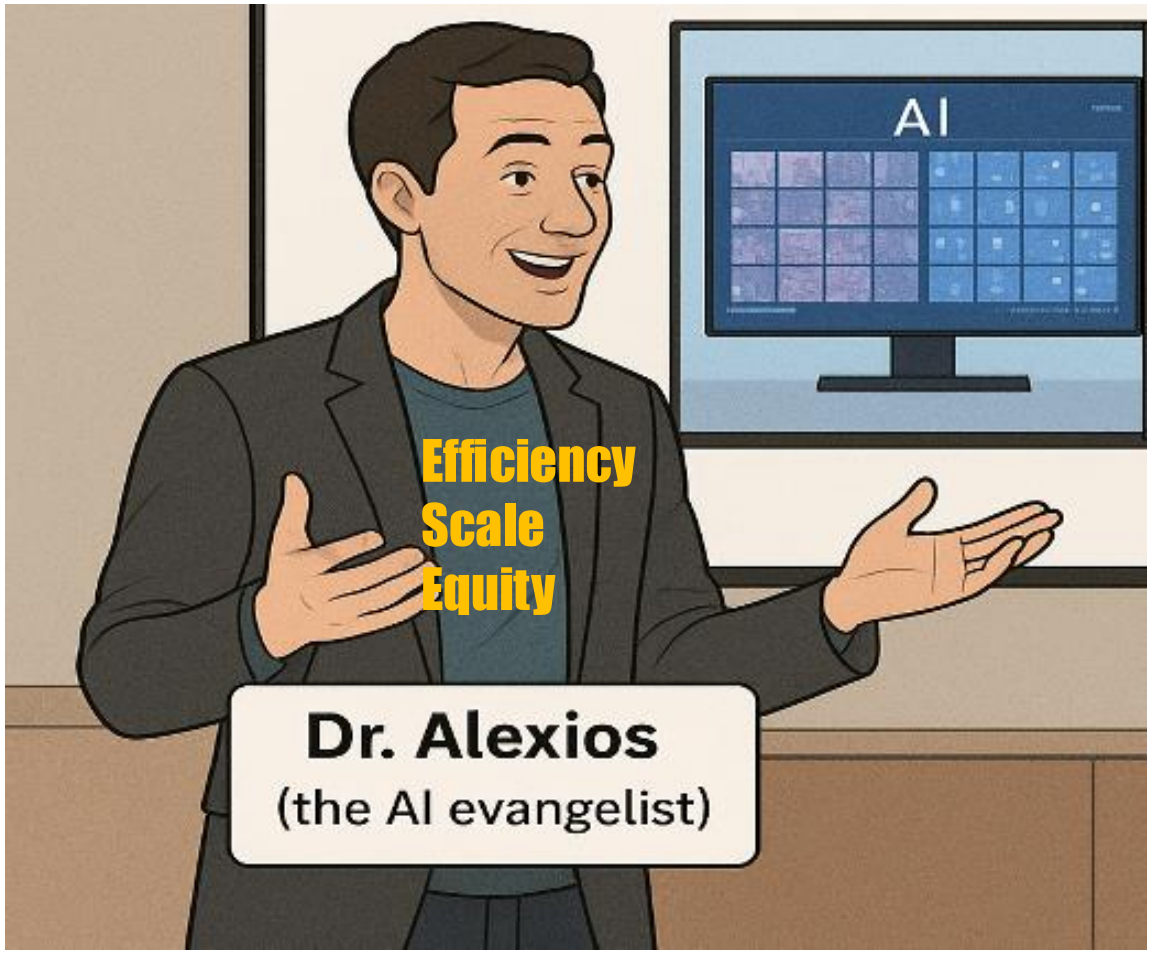
In 2021, WHO published comprehensive guidance on the ethics and governance of AI for health, developed through consultation with 20 AI experts. The report outlined potential benefits and risks of AI in healthcare and proposed six consensus-based principles to guide governments, developers, and providers.



1. Protect autonomy
2. Promote human well-being, human safety and the public interest
3. Ensure transparency, explainability and intelligibility
4. Foster responsibility and accountability
5. Ensure inclusiveness and equity
6. Promote AI that is responsive and sustainable

World Health Organization. (2021). *Ethics and governance of artificial intelligence for health: WHO guidance.* Geneva: World Health Organization. https://www.who.int/publications/i/item/9789240029200

AI in Lab Medicine and Pathology





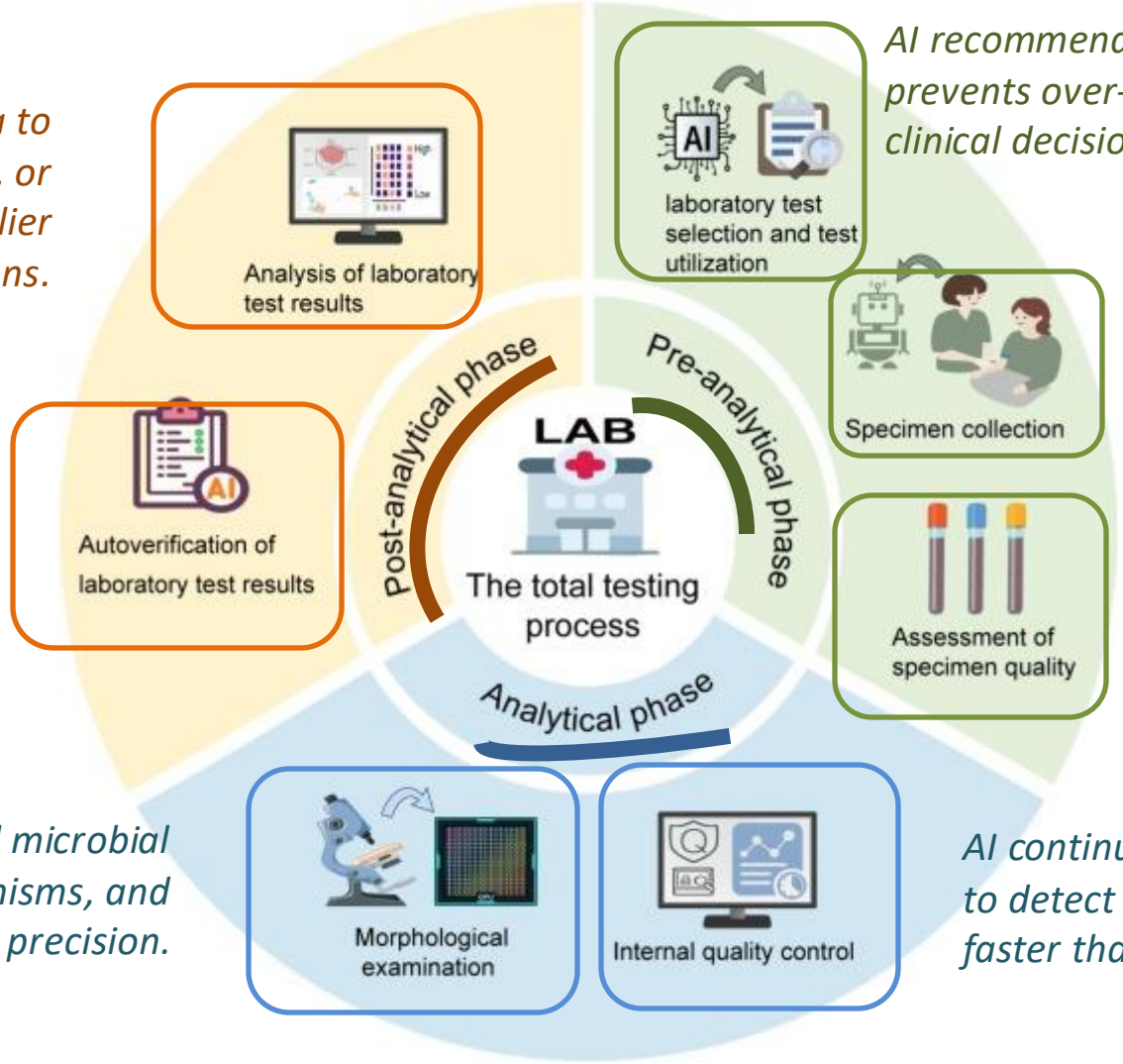
both of these protagonists have an important insight; but it is incorrect to think that AI technology will only have one (either-or) effect

AI in Lab Medicine: Opportunities

AI interprets complex lab data to predict risks—such as sepsis, or anemia—enabling earlier personalized interventions.

AI autonomously validates results, escalating only atypical cases for human review.

AI analyzes blood, urine, and microbial images to classify cells, organisms, and disease subtypes with near-expert precision.



AI recommends appropriate laboratory tests and prevents over- or under-ordering, optimizing clinical decisions and resource use.

AI-assisted systems use imaging and robotics to identify veins and perform safer, more accurate blood draws.

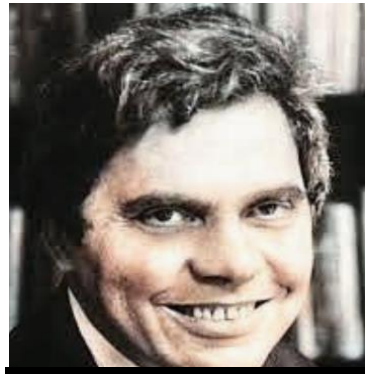
AI detects mislabeled, degraded, or mismatched specimens early, minimizing pre-analytical errors.

AI continuously monitors data trends to detect analytical drifts or errors faster than traditional QC methods.

Hou, H., Zhang, R., & Li, J. (2024). Artificial intelligence in the clinical laboratory. *Clinica Chimica Acta*, *559*, 119724.

AI in Lab Medicine: Both-And Multiplex Thinking


Melvin Kranzberg


Neil Postman

Kranzberg's First Law of Technology

"Technology is neither good nor bad;
nor is it neutral."

"Every technology is
both a burden and a blessing;
not either-or, but this-and-that."

Instead of Either-Or:

Think

—Neither-Nor

—Both-And

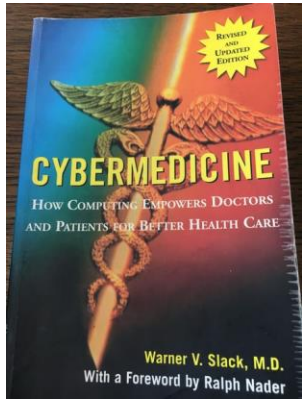


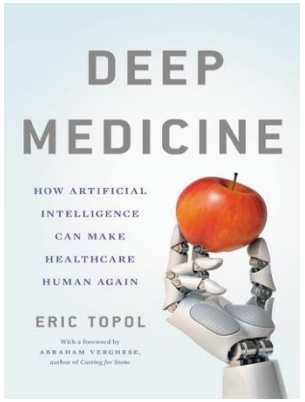
For most UN SDGs, AI, paradoxically, can act both as an enabler and an inhibitor.

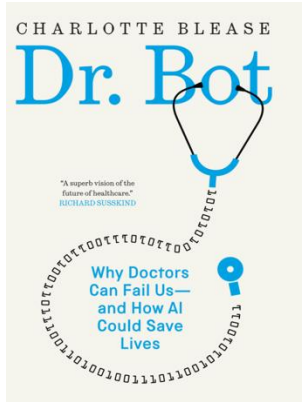
Vinuesa, Ricardo et al., "The role of artificial intelligence in achieving the Sustainable Development Goals." Nature communications 11, no. 1 (2020): 1-10.

How to have the best of both worlds?

Technology-Centered Efficiency **AND** **Human-Centered Quality of Service**







What if technology could help us enhance our humanity?



OTHER POLARITIES

Codes of Ethics **AND** **Regulation/ Law**

For more context, please read: **Qadir, J.**, Qoronfleh, M. W., & Senturk, R. (2025). Toward multiplex health: integrating complexity, normativity, and Open Science. *Frontiers in Psychology*, *16*, 1603474.

Defining the Boundaries of Human and AI

***"Render unto man the things which are man's and unto the computer the things which are the computer's. This would seem the intelligent policy to adopt when we employ men and computers together."*—Wiener, 1964**

Human driver monitors environment			System monitors environment		
0	1	2	3	4	5
No automation	Driver assistance	Partial automation	Conditional automation	High automation	Full automation



0 – Human doctors in full control.

1–2 – Increasing automation
Machines assist doctors with decision support.

3 – Conditional automation
Machines take on more monitoring and analysis, humans provide oversight.

4 – High automation (unlikely)
AI handles nearly all decisions; humans intervene rarely.

5 – Full automation
AI-only doctor with no human oversight — unrealistic/ unlikely.

OVERSIGHT OF THE ENTIRE HEALTHCARE AI LIFECYCLE.



npj | digital medicine — Perspective

Published in partnership with Seoul National University Bundang Hospital

https://doi.org/10.1038/s41746-025-01614-1

A "True Lifecycle Approach" towards governing healthcare AI with the GCC as a global governance model

Check for updates

Barry Solaiman[1,2], Yosra Magdi Mekki[3,4], Junaid Qadir[3], Mohammed Ghaly[1], Mohamed Abdelkareem[3,4] & Abdulla Al-Ansari[3,4]



bioethics

SPECIAL ISSUE ARTICLE

Assisting, Replicating, or Autonomously Acting? An Ethical Framework for Integrating AI Tools and Technologies in Healthcare

Aasim I. Padela, Rwan Hayek, Aliya Tabassum, Fabrice Jotterand, Junaid Qadir

First published: 18 July 2025 | https://doi.org/10.1111/bioe.70019

Ethically and logistically problematic

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Ethical use of AI in healthcare demands balanced judgment, technological literacy, and a systems approach—cultivating both–and wisdom that harmonizes innovation with integrity, efficiency with empathy, and automation with accountability.

THANK YOU!

Comments/ Questions?

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